Let $f(x) \in F[x]$ be given with the irreducible factorization

$$f(x) = f_1(x) \cdots f_n(x).$$

Since every Automorphism fixing F maps root of f_i to the root of f_i, the Same irreducible Polynomial, therefore, for the Splitting field K of f,

$$\mathrm{Aut}(K/F) \hookrightarrow S_{f_1} \times \cdots \times S_{f_n}.$$

With such an observation, in this paper, we figure out concretely the Galois group of the Polynomials.

Def. Let $x_1, \cdots, x_n$ be indeterminates.
The elementary symmetric functions $s_1, s_2, \cdots, s_n$ are defined by

$$s_1 = x_1 + x_2 + \cdots + x_n$$
$$s_2 = x_1x_2 + x_1x_3 + \cdots + x_1x_n + x_2x_3 + x_2x_4 + \cdots + x_{n-1}x_n$$
$$\vdots$$
$$s_n = x_1x_2 \cdots x_n$$

And, the general polynomial of degree n is the polynomial

$$f(x) = (x-x_1)(x-x_2)\cdots(x-x_n).$$

Note that

$$f(x) = x^n - s_1x^{n-1} + s_2x^{n-2} - \cdots + (-1)^n \cdot s_n$$

Thm. Let $x_1, \cdots, x_n$ be indeterminates of a field F, and $s_1, \cdots, s_n$ be symmetric functions. Then, $F(x_1, \cdots, x_n)$ is Galois extension of $F(s_1, \cdots, s_n)$ with

$$\mathrm{Gal}\left(F(x_1, \cdots, x_n)/F(s_1, \cdots, s_n)\right) \cong S_n.$$

Proof. Since $F(x_1, \cdots, x_n)$ is the splitting field of the polynomial

$$f(x) = (x-x_1)\cdots(x-x_n) = x^n - s_1x^{n-1} + \cdots + (-1)^n \cdot s_n$$

over $F(s_1, \cdots, s_n)$ and f is separable, so the first assertion is obtained.
And, since degree of f is n, so $[F(x_1, \cdots, x_n) : F(s_1, \cdots, s_n)] \leq n!$.
Meanwhile, for any permutation $\sigma \in S_n$, a induced map

$$\bar{\sigma} : F(x_1, \cdots, x_n) \to F(x_1, \cdots, x_n) : \frac{g(x_1, \cdots, x_n)}{f(x_1, \cdots, x_n)} \mapsto \frac{g(x_{\sigma(1)}, \cdots, x_{\sigma(n)})}{f(x_{\sigma(1)}, \cdots, x_{\sigma(n)})}$$

is an Automorphism which fixes $F(s_1, \cdots, s_n)$, so

$$S_n \hookrightarrow \mathrm{Aut}\left(F(x_1, \cdots, x_n)/F(s_1, \cdots, s_n)\right)$$

Since $|S_n| = n!$, so proved. ▣